MIL-H-639D
30 September 1986
SUPERSEDING
MIL-H-639C
21 April 1969

MILITARY SPECIFICATION

HOOKS (FOR BELTS, STRAPS, AND EQUIPAGE ITEMS)

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This document covers metal hooks for belts, straps, and equipage items.

1.2 Classification. Hooks shall be of the following types, classes, and sizes as specified (see 6.2):

Type I - Hook, double, vertical loop

Class 1 - Brass wire
Class 2 - Steel wire

Sizes 1 and 2 (see Drawing 4-1-67)

Type II - Hook, double, horizontal loop

Class 1 - Brass wire
Class 2 - Steel wire

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to U.S. Army Natick Research, Development, and Engineering Center, Natick, MA 01760-5014, by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

AMSC N/A FSC 5340

DISTRIBUTION STATEMENT A. Approved for public release; distribution is unlimited.

Source: https://assist.dla.mil -- Downloaded: 2026-10-16T10:59Z
Check the source to verify that this is the current version before use.

Type III - Hook, hanger, canteen

Class 1 - Brass wire
Class 2 - Steel wire

Type IV - Hook, end, double

Class 1 - Brass wire
Class 2 - Steel wire

Type V - Hook, loop

Class 1 - Brass wire
Class 2 - Steel wire

Type VI - Hook, back strap, with roller, 1 1/4 inch

Type VII - Hook, side strap, with roller, 1 inch

Type VIII - Hook, "S"

Type IX - Hook, end

Type X - Hook, stud

Class 1 - Brass wire
Class 2 - Iron, malleable

Type XI - Hook, sliding, 5/8 inch

Class 1 - Brass wire
Class 2 - Steel

Type XII - Hook, "S" (commercial)

Class 1 - Brass wire

Sizes A to K (see Drawing 4-1-67)

Class 2 - Steel wire

Sizes A to K (see Drawing 4-1-67)

Source: https://assist.dla.mil -- Downloaded: 2026-10-16T10:59Z
Check the source to verify that this is the current version before use.

2. APPLICABLE DOCUMENTS

2.1 Government documents.

* 2.1.1 Documents. The following documents form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents shall be those listed in the issue of the Department of Defense Index of Specifications and Standards (DODISS) and supplement thereto, cited in the solicitation.

SPECIFICATIONS

FEDERAL

QQ-B-613	- Brass, Leaded and Nonleaded: Flat Products (Plate, Bar, Sheet, and Strip)
QQ-C-390	- Copper Alloy Castings (Including Cast Bar)
QQ-I-666	- Iron Castings, Malleable
QQ-N-290	- Nickel Plating (Electrodeposited)
QQ-S-698	- Steel Sheet and Strip, Low Carbon
QQ-W-321	- Wire, Copper Alloy
QQ-W-461	- Wire, Steel, Carbon (Round, Bare and Coated)
TT-E-529	- Enamel, Alkyd, Semigloss (Type II)
PPP-B-601	- Boxes, Wood, Cleated-Plywood
PPP-B-636	- Boxes, Shipping, Fiberboard

MILITARY

MIL-F-495	- Finish, Chemical, Black, for Copper Alloys

STANDARDS

MILITARY

MIL-STD-105	- Sampling Procedures and Tables for Inspection by Attributes
MIL-STD-129	- Marking for Shipment and Storage
MIL-STD-147	- Palletized Unit Loads

DRAWINGS

U.S. ARMY NATICK RESEARCH, DEVELOPMENT, AND ENGINEERING CENTER

4-1-67	- Hooks, for Belts, Straps and Equipage Items
4-1-68	- Hooks, for Belts, Straps and Equipage Items

Source: https://assist.dla.mil -- Downloaded: 2026-10-16T10:59Z
Check the source to verify that this is the current version before use.

(Copies of documents required by contractors in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting activity.)

* 2.2 Other publications. The following documents form a part of this document to the extent specified herein. Unless otherwise specified, the issues of the documents which are DOD adopted shall be those listed in the issue of the DODISS specified in the solicitation. Unless otherwise specified, the issues of documents not listed in the DODISS shall be the issues of the nongovernment documents which are current on the date of the solicitation.

AMERICAN SOCIETY FOR TESTING AND MATERIALS (ASTM)

B 633 - Electrodeposited Coatings of Zinc on Iron and Steel

D 3951 - Standard Practice for Commercial Packaging

(Application for copies should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, PA 19103.)

(Technical society and technical association documents are generally available for reference from libraries. They are also distributed among technical groups and using Federal agencies.)

* 2.3 Order of precedence. In the event of conflict between the text of this document and the references cited herein, the text of this document shall take precedence. Nothing in this document, however, shall supersede applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 First article. When specified in the contract or purchase order, a sample shall be subject to first article inspection (see 4.3, 6.2, and 6.3).

* 3.2 Materials. The specified materials shall conform to the referenced documents specified herein. Recycled material may be used (see 6.4).

3.2.1 Brass casting. The brass casting shall conform to alloy No. A1 or A3 of QQ-C-390.

3.2.2 Brass wire. The brass wire shall conform to alloy No. 260 or 270, three-quarter hard temper as specified in QQ-W-321.

3.2.3 Brass sheet or strip. The brass sheet or strip shall conform to alloy No. 260 or alloy No. 268, half-hard temper of QQ-B-613.

Source: https://assist.dla.mil -- Downloaded: 2026-10-16T10:59Z
Check the source to verify that this is the current version before use.

3.2.4 Steel wire. The steel wire shall conform to steel No. 1015 or 1035, finish 1, hard temper of QQ-W-461.

3.2.5 Steel sheet or strip. The steel sheet shall conform to hot rolled commercial quality (HRCQ) or cold rolled commercial quality (CRCQ) of QQ-S-698. The steel strip shall conform to hot rolled commercial quality or cold rolled half-hard, No. 2 temper of QQ-S-698.

3.2.6 Iron, malleable. The malleable iron castings shall conform to grade I (35018) or grade IG (35018G) of QQ-I-666, except that marking requirements do not apply.

* 3.2.7 Enamel. The enamel shall conform to TT-E-529. The color shall be black.

3.3 Construction. The construction of the hooks shall conform to the design, dimensions, and sizes as shown on Drawings 4-1-67 and 4-1-68.

3.3.1 Types I, II, III, IV, V, and XII, class 1 hooks. The types I, II, III, IV, V, and XII, class 1 hooks shall be fabricated from material specified in 3.2.2. The type III, class 1 and type V, class 1 hooks shall have the ends joined after forming by brazing. The joint shall be sound and smooth.

3.3.2 Types I, II, III, IV, V, and XII, class 2; and types VIII and IX hooks. The types I, II, III, IV, V, and XII, class 2, and types VIII and IX hooks shall be fabricated from material specified in 3.2.4, except that for type I, size 2 only, the steel number shall be 1035. The type III, class 2, and type V, class 2 hooks shall have the ends joined after forming by welding. The welded joints shall be sound and smooth.

3.3.3 Types VI and VII hooks. The types VI and VII hooks shall be cast from grade I material specified in 3.2.6.

3.3.3.1 Rollers for types VI and VII hooks. The rollers for the hooks shall be fabricated from materials specified in 3.2.5. The rollers shall rotate freely, and the butting edges shall be smooth and free from burrs or rough edges.

3.3.4 Type X, class 1 hook. The type X, class 1 hook shall be cast from material specified in 3.2.1.

3.3.5 Type X, class 2 hook. The type X, class 2 hook shall be cast from grade IG material specified in 3.2.6.

Source: https://assist.dla.mil -- Downloaded: 2026-10-16T10:59Z
Check the source to verify that this is the current version before use.

3.3.6 Type XI, class 1 hook. The type XI, class 1 hook shall be fabricated from material specified in 3.2.3 and to the dimensions specified on Drawing 4-1-68.

3.3.7 Type XI, class 2 hook. The type XI, class 2 hook shall be fabricated from material specified in 3.2.5 and to the dimensions specified on Drawing 4-1-68.

3.3.8 Castings. All castings shall be sound and free from pits, blowholes, porosity, fractures, and cracks. The castings shall be smooth and clean prior to application of finish.

3.3.9 Stampings. All fabricated stampings shall be free from rough and sharp edges before final finish is applied.

3.4 Finish.

3.4.1 Types I, II, III, X, and XI, class 1 hooks. The types I, II, III, X, and XI, class 1 hooks shall be given a black chemical finish and shall show resistance to hot soap solution and resistance to accelerated weathering in accordance with MIL-F-495.

* 3.4.2 Types I, II, III, IV, V, X, and XI, class 2; and types VI, VII, VIII and IX hooks. The types I, II, III, IV, V, X, and XI, class 2, and types VI, VII, VIII, and IX hooks shall be given a phosphate-treated zinc plate conforming to type IV, classification Fe/Zn13 Sc 3 of ASTM B 633, followed by baked enamel as specified in 3.2.7. The coating shall level out to a uniform dry film thickness without orange peel, wrinkles, drops, streaks, or areas of no film.

3.4.3 Types IV and V, class 1 hooks. The types IV and V, class 1 hooks shall be given a bright nickel finish conforming to type V, class 1 of QQ-N-290. Finished hooks shall be buffed and polished.

3.4.4 Type XII, class 1 hooks. The type XII, class 1 brass hooks shall have a buff bright brass finish.

3.4.5 Type XII, class 2 hooks. The type XII, class 2 steel hooks after complete fabrication shall be galvanized by the hot dip method. The galvanizing shall be in accordance with the manufacturer's commercial practice, uniformly smooth, adherent, and free from drippings.

3.5 Marking. All hooks shall be marked in an inconspicuous location with the manufacturer's identification in such a manner that the stamping is not distorted, bent, or malformed.

Source: https://assist.dla.mil -- Downloaded: 2026-10-16T10:59Z
Check the source to verify that this is the current version before use.

3.6 Workmanship. All surfaces shall be smooth and free from burrs, slivers, rough or sharp edges, pits, blowholes, porosity, and cracks.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in this document where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

* 4.1.1 Responsibility for compliance. All items must meet all requirements of sections 3 and 5. The inspection set forth in this document shall become a part of the contractor's overall inspection system or quality program. The absence of any inspection requirement in the document shall not relieve the contractor of the responsibility of assuring that all products or supplies submitted to the Government for acceptance comply with all requirements of the contract. Sampling in quality conformance does not authorize submission of known defective material, either indicated or actual, nor does it commit the Government to acceptance of defective material.

4.2 Classification of inspections. The inspection requirements specified herein are classified as follows:

a. First article inspection (see 4.3).
b. Quality conformance inspection (see 4.4).

4.3 First article inspection. When a first article is required (see 3.1), it shall be examined for the defects specified in 4.4.3 and 4.4.4. Types I, II, III, X, and XI, class 1 hooks shall be tested for the characteristics specified in 4.4.5. The presence of any defect or failure to pass any test, when applicable, shall be cause for rejection of the first article.

4.4 Quality conformance inspection. Unless otherwise specified, sampling for inspection shall be performed in accordance with MIL-STD-105.

4.4.1 Component and material inspection. In accordance with 4.1, components and materials shall be inspected in accordance with all the requirements of referenced documents unless otherwise excluded, amended, modified, or qualified in this document or applicable purchase document.

Source: https://assist.dla.mil -- Downloaded: 2026-10-16T10:59Z
Check the source to verify that this is the current version before use.

4.4.2 In-process inspection. Inspection of sub-assemblies shall be made to ascertain that construction details which cannot be examined in the finished product are in accordance with specified requirements. The Government reserves the right to exclude from consideration for acceptance, any material or service for which in-process inspection has indicated nonconformance.

4.4.2.1 Plating. Examination shall be conducted for application of plating specified in 3.4.2, prior to application of enamel. Whenever nonconformance is noted, correction shall be made to affected items and process.

* 4.4.3 End item visual examination. The end item shall be examined for the defects listed in table I. The lot size shall be expressed in units of hooks of one type, class, and size. The sample unit shall be one completely fabricated hook. The inspection level shall be II and the acceptable quality level (AQL), expressed in terms of defects per hundred units, shall be 2.5 for major defects and 6.5 for total defects.

TABLE I. End item visual defects

Examine	Defect	Classification	
		Major	Minor
Finish	Not smooth		201
	Not color specified, when applicable		202
	Not clean (i.e., dirt, dust, grease stain, oil, or welding flux embedded in finish)		203
	Nickel finish hooks not buffed and polished		204
	Enamel finished hooks have orange peel, wrinkles, drops, streaks, or areas of no film		205
	Hot dipped galvanized hooks not uniformly smooth, adherent, or free from drippings		206
	Type XII, class 1 hooks not bright finished		207
Construction and workmanship	Size or design not as specified	101	
	Unsound castings (i.e., porous, warped, cracked, fractured, or contain blowholes)	102	

Source: https://assist.dla.mil -- Downloaded: 2026-10-16T10:59Z
Check the source to verify that this is the current version before use.

TABLE I. End item visual defects

Examine	Defect	Classification	
		Major	Minor
Construction and workmanship - continued	Stamping distorted, bent, or malformed	103	
	Rough or sharp edges not removed	104	
	Weld or braze missing, broken, or not fused, smooth, or sound, where applicable	105	
	Rollers will not rotate freely	106	
Marking	Missing, incorrect, incomplete, or illegible		208

4.4.4 End item dimensional examination. The end item shall be examined for the dimensions specified on Drawings 4-1-67 and 4-1-68. The lot size shall be expressed in units of hooks of one type, class, and size. The sample unit shall be one completely fabricated hook. The inspection level shall be S-2 and the AQL, expressed in terms of defects per hundred units, shall be 4.0.

4.4.5 End item testing. The types I, II, III, X, and XI, class 1 hooks shall be tested for the characteristics in table II. The lot size shall be expressed in units of hooks. The sample unit shall be three hooks, one for each test. The inspection level shall be S-1 and the AQL, expressed in terms of defects per hundred units, shall be 6.5.

Source: https://assist.dla.mil -- Downloaded: 2026-10-16T10:59Z
Check the source to verify that this is the current version before use.

TABLE II. End item tests

Characteristics	Requirement paragraph	Test method	No. of determinations per sample unit	Results reported as
Black chemical finish Color and gloss	3.4.1	MIL-F-495	1	Pass or fail
Resistance to hot soap solution	3.4.1	MIL-F-495	1	Pass or fail
Resistance to accelerated weathering	3.4.1	MIL-F-495	1	Pass or fail

4.4.6 Packaging examination. The fully packaged end items shall be examined for the defects listed below. The lot size shall be expressed in units of shipping containers. The sample unit shall be one shipping container fully packaged. The inspection level shall be S-2 and the AQL, expressed in terms of defects per hundred units, shall be 2.5.

Examine	Defect
Marking (exterior and interior)	Omitted; incorrect; illegible; of improper size, location, sequence, or method of application
Materials	Any component missing, damaged, or not as specified

Source: https://assist.dla.mil -- Downloaded: 2026-10-16T10:59Z
Check the source to verify that this is the current version before use.

Examine	Defect
Workmanship	Inadequate application of components, such as incomplete sealing or closure of flap, improper taping, loose strapping, or inadequate stapling Bulged or distorted container
Content	Net weight per container is more or less than required

4.4.7 Palletization examination. The fully packaged and palletized end items shall be examined for the defects listed below. The lot size shall be expressed in units of palletized unit loads. The sample unit shall be one palletized unit load, fully packaged. The inspection level shall be S-1 and the AQL, expressed in terms of defects per hundred units, shall be 6.5.

Examine	Defect
Finished dimensions	Length, width, or height exceeds specified maximum requirement
Palletization	Pallet pattern not as specified Interlocking of loads not as specified Load not bonded with required straps as specified
Weight	Exceeds maximum load limits
Marking	Omitted; incorrect; illegible; of improper size, location, sequence, or method of application

5. PACKAGING

5.1 Preservation. Preservation shall be level A or Commercial, as specified (see 6.2).

5.1.1 Level A. Twenty pounds of hooks of one type, class, and size only shall be unit packed in a snug-fitting fiberboard box conforming to style RSC, W5c, or W5s of PPP-B-636. Each box shall be agitated from time to time to ensure a compact and well filled box. Each box shall be closed in accordance with the appendix of PPP-B-636.

Source: https://assist.dla.mil -- Downloaded: 2026-10-16T10:59Z
Check the source to verify that this is the current version before use.

* 5.1.2 Commercial. Hooks shall be preserved in accordance with ASTM D 3951.

5.2 Packing. Packing shall be level A, B, or Commercial, as specified (see 6.2).

* 5.2.1 Level A packing. One hundred and twenty pounds of hooks of one type, class, and size only, preserved as specified in 5.1, shall be packed in a snug-fitting box conforming to overseas type of PPP-B-601, style optional.

5.2.2 Level B packing. One hundred and twenty pounds of hooks of one type, class, and size only, preserved as specified in 5.1, shall be packed in a snug-fitting, double-walled, fiberboard shipping container conforming to style FTC, type CF, class domestic of PPP-B-636. Each shipping container shall be closed and reinforced with flat strapping or tape banding in accordance with the appendix of PPP-B-636.

5.2.2.1 Weather-resistant shipping containers. When specified (see 6.2), the shipping container shall be a grade V3, V3s, or V4s fiberboard box fabricated in accordance with PPP-B-636 and closed in accordance with the appendix of PPP-B-636.

* 5.2.3 Commercial packing. Hooks, preserved as specified in 5.1, shall be packed in accordance with ASTM D 3951.

* 5.3 Palletization. When specified (see 6.2), hooks, packed as specified in 5.2, shall be palletized on a 4-way entry pallet in accordance with load type I or Ia, as applicable, of MIL-STD-147. Pallet type shall be type I (4-way entry), type IV or type V in accordance with MIL-STD-147. Each prepared load shall be bonded with primary and secondary straps in accordance with bonding means K and L or film bonding means O or P. Pallet pattern shall be in accordance with appendix of MIL-STD-147. Interlocking of loads shall be effected by reversing the pattern of each course. If the container is of a size that does not conform to any of the pallet patterns specified in MIL-STD-147, the pallet patterns used shall first be approved by the contracting officer.

* 5.4 Marking. In addition to any special marking required by the contract or purchase order, unit packs, shipping containers, and palletized unit loads shall be marked in accordance with MIL-STD-129 or ASTM D 3951, as applicable. Each interior package and shipping container shall be marked with the approximate quantity of items contained therein.

Source: https://assist.dla.mil -- Downloaded: 2026-10-16T10:59Z
Check the source to verify that this is the current version before use.

6. NOTES

6.1 Intended use. Hooks are intended to be used with belts, straps, and equipage items used by personnel of the Armed Forces.

* 6.2 Ordering data. Acquisition documents should specify the following:

a. Title, number, and date of this document.
b. Type, class, and size hook required (see 1.2).
c. When a first article is required (see 3.1, 4.3, and 6.3).
d. Selection of applicable levels of preservation and packing (see 5.1 and 5.2).
e. When weather-resistant fiberboard boxes are required for level B shipment (see 5.2.2.1).
f. When palletization is required (see 5.3).

* 6.3 First article. When a first article is required, it shall be inspected and approved under the appropriate provisions of FAR 52.209. The first article should be a preproduction sample. The contracting officer should specify the appropriate type of first article and the number of units to be furnished. The contracting officer should include specific instructions in all acquisition instruments regarding arrangements for selection, inspection, and approval of the first article.

* 6.4 Recycled material. It is encouraged that recycled material be used when practical as long as it meets the requirements of this document (see 3.2).

6.5 Subject term (key word) listing.

Fasteners
Hooks

6.6 Changes from previous issue. The margins of this document have been marked with an asterisk (*) to indicate where changes (additions, modifications, corrections, deletions) from the previous issue were made. This was done as a convenience only, and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content, as written, irrespective of the marginal notations and relationship to the last previous issue.

Source: https://assist.dla.mil -- Downloaded: 2026-10-16T10:59Z
Check the source to verify that this is the current version before use.

Custodians:

Army - GL
Navy - YD
Air Force - 99

Review activities:

Air Force - 82
DLA - IS

User activities:

Army - ME
Navy - MC

Preparing activity:

Army - GL

Project No. 5340-1687

Source: https://assist.dla.mil -- Downloaded: 2026-10-16T10:59Z
Check the source to verify that this is the current version before use.

INSTRUCTIONS: In a continuing effort to make our standardization documents better, the DoD provides this form for use in submitting comments and suggestions for improvements. All users of military standardization documents are invited to provide suggestions. This form may be detached, folded along the lines indicated, taped along the loose edge *(DO NOT STAPLE)*, and mailed. In block 5, be as specific as possible about particular problem areas such as wording which required interpretation, was too rigid, restrictive, loose, ambiguous, or was incompatible, and give proposed wording changes which would alleviate the problems. Enter in block 6 any remarks not related to a specific paragraph of the document. If block 7 is filled out, an acknowledgement will be mailed to you within 30 days to let you know that your comments were received and are being considered.

NOTE. This form may not be used to request copies of documents, nor to request waivers, deviations, or clarification of specification requirements on current contracts. Comments submitted on this form do not constitute or imply authorization to waive any portion of the referenced document*(s)* or to amend contractual requirements.

(Fold along this line)

(Fold along this line)

DEPARTMENT OF THE ARMY
Army Natick Research and Development Center
Natick, Massachusetts 01760

NO POSTAGE NECESSARY IF MAILED IN THE UNITED STATES

OFFICIAL BUSINESS
PENALTY FOR PRIVATE USE $300
STRNC-ES

BUSINESS REPLY MAIL
FIRST CLASS PERMIT NO 12062 WASHINGTON D C
POSTAGE WILL BE PAID BY THE DEPARTMENT OF THE ARMY

Commander
US Army Natick Research and Development Center
ATTN: STRNC-ES
Natick, MA 01760

Source: https://assist.dla.mil -- Downloaded: 2026-10-16T10:59Z
Check the source to verify that this is the current version before use.

STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

(See Instructions – Reverse Side)

1. DOCUMENT NUMBER	2 DOCUMENT TITLE
MIL-H-639D	Hooks, (For Belts, Straps, and Equipage Items)

3a. NAME OF SUBMITTING ORGANIZATION	4 TYPE OF ORGANIZATION *(Mark one)*
	☐ VENDOR
	☐ USER
b ADDRESS *(Street, City, State, ZIP Code)*	☐ MANUFACTURER
	☐ OTHER *(Specify)*. _____

5 PROBLEM AREAS

a. Paragraph Number and Wording

b Recommended Wording.

c Reason/Rationale for Recommendation

6 REMARKS

7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional	b. WORK TELEPHONE NUMBER *(Include Area Code)* – Optional
c. MAILING ADDRESS *(Street, City, State, ZIP Code)* – Optional	8. DATE OF SUBMISSION *(YYMMDD)*

DD FORM 1426

Source: https://assist.dla.mil – Downloaded: 2026-10-16T10:50Z
Check the source to verify that this is the current version before use.